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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Quintuplets
Project / Solution Name	Waitly – Ordering App
Name (Leader)	Dias Zhumadilda
Email & Phone (Leader)	Cys2509080@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://waitly.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Dias-dev-art/Waitly_Ordering-App.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A customer sends a message to the shop's Telegram bot (1 Telegram bot and 1 link for each shop). The message is send to a webhook on our FastAPI backend that is hosted on a Replit Reserv. VM: The backend checks the sender, puts the message in the database of PostgreSQL and generates a system prompt from the shop's menu and the current cart. This is sent to a Qwen2.5-7B-Instruct endpoint on RunPod GPU. Through an OpenAI compatible API supported by tool calling. The model responds with text or function calls. The tool call is checked on the database whether it exists, for a user or group with access, or for an item that is not available. The total is calculated in Python, and then order is updated, the list of items available on the menu are computed in Python. To prevent such content from being returned to the Telegram Bot API, reply guard applies filters: the shop's DuitNow QR image. The owner logs into the Next.js dashboard, on the same FastAPI process, and gets live order data. Approve or Not will invoke a pre-made confirmation template to the customer. No unpaid orders are cancelled after 10 minutes for timers that are in the same process. Secrets are kept in Replit Secrets: The LLM endpoint is changed through configuration only.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Backend Language	PSF License
FastAPI 0.158.0 (with Uvicorn 0.54.0, Starlette 1.9.0)	REST API, Telegram webhook, SSE	MIT/BSD-3-Clause
SQLModel 0.0.48, Pydantic 2.14.1, pydantic-settings 2.16.1	Data Models, validation, configuration	MIT
Alembic 1.20.0, psycopg 3.3.6	Database migration and PostgreSQL driver	MIT/LGPL-3.0
Openai (Python SDK) 3.36.1	OpenAI-compatible client for the RunPod LLM endpoint	Apache-2.0

Name & Version	Purpose in Project	License
httpx 0.28.1	HTTP client for the messaging API	BSD-3-Clause
PyJWT 2.15.1, pwdlib 0.3.1 (argon2-cffi 25.1.0, bcrypt 5.0.0)	Password hashing, owner login tokens	MIT/Apache-2.0
cryptography 50.0.2	Fernet encryption for each shop access tokens	Apache-2.0/BSD-3-Clause
pytest 9.1.1, pytest-asyncio 1.4.0, ruff 0.17.1 (dev only)	Automated test and linting	MIT
TypeScript 5.9.3	Frontend language	Apache-2.0
Next.js 16.4.2, React 19.3.0, React DOM 19.3.0	Owner dashboard	MIT
Tailwind CSS 4.3.3, tw-animate-css 1.4.0	Styling	MIT
shadcn/ui 4.22.1, Radix UI 1.8.1, class-variance-authority 0.7.1	Accessible UI components	MIT/Apache-2.0
TanStack React Query 5.104.5	Data fetching, caching	MIT
React Hook Form 7.90.0, Zod 4.6.7, @hookform/resolvers 5.9.2	Forms and validation	MIT
Lucide-react 1.16.0, sonner 2.0.8, next-themes 0.4.6	Icons, theming	ISC-MIT
PostgreSQL 16	Relational database	PostgreSQL License
Qwen2.5-7B	AI model	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Telegram Bot API	Customer chat channel: receives messages by webhook, replies, DuitNow QR images, fixed confirmations	Free
Open-AI-compatible Chat Completion API on Runpod	LLM inference with tool calling order agent (Qwen2.5-7B-instruct)	Sponcor-provided(Runpod), Free (AI model)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit PostgreSQL database	Stores menus, order, chats, businesses and screenshots of QR payment	Replit-managed PostgreSQL
Docker Compose and ngrok (local development)	Local PostgreSQL 16, public tunnel for webhook testing, not used in production	Free, local machine

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2.5-7B-Instruct	Alibaba Cloud team, operating on Runpod	Fine-tuned	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo shop seed data	Repository: Src/backend/app/seed.py	Synthetic	Apache-2.0	N
Evaluation scenarios	src/backend/app/evals/llm_scenarios.py	Synthetic	Apache-2.0	N
Runtime shop, customer data	Pinterest	Public	User content	Y
Fine-tuning	Replit	Synthetic	Apache-2.0	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude/Claude Code	Reviewing and drafting frontend and backend code	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All code generated by the AI was considered untrusted. It was added to the repository via pull requests. The team members reviewed it and it needed to pass the automated test suite (which was over 100 tests, and a scripted fake LLM). The latter was used with and without the ruff linter and was manually executed in the dev simulator. Prices, totals, order – critical logic , he team's components are explicitly specified in plain Python (not relying on the model), and the status (payment wording) all code, text and design submitted will remain the property of remain.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits, platform	Reserved VM deployment of the entire prototype: API, Telegram webhook, dashboard, database
Runpod	Platform, API, credits	Serves Qwen2.5-7B-Instruct, every customer text messages as a request to assist them

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Both are the platforms that the prototype runs on but they are not hard dependencies. The LLM is only selected by the configuration (base URL, key, model name), which means that RunPod can be replaced with any other service. We have previously designed for OpenRouter; it is an openAI compatible provider. The back end is a typical FastAPI application with PostgreSQL database and a Docker Compose configuration, which can be substituted with any other application/container. A single process that is long-lived and runs on host. RunPod is key in the demo as it will be the host of the model that powers the agent.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Collected: Telegram identifier and display name of customers, messages in the chat, delivery address, order details. Details and optional payment photos; owners' email, name and hashed password; shop menu and DuitNow QR. Sent to the AI: Only the chat text, shop name, menu, current cart and address, sent to our own RunPod No customer identifiers or owner credentials are ever included, only endpoint (not a public AI service).
Data Separation: Per-shop data separation, Argon2/bcrypt These features include password hashing, encrypted shop tokens, upload file size limits and logs that Exclude tokens and entire phone numbers. Retention: webhook de-duplication IDs expire after 7 days, orders and chats are retained until the operator deletes them. An automatic policy for retention is planned.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Model errors: a Qwen-7B model might misread intent, or get into the wrong language, or simply write out a tool name as plain.
Mitigation: each tool call is checked with the database and totals are calculated in code and a reply guard is used to prevent multiple entries from appearing. text-tool recovery step runs the intended tool, a blocks tool name and "payment received" claim. timeout of 60 s+ with the latency of 1.5–7 s per turn (occasionally 40 s+ with cold GPU workers). Retries and a warm worker for demos.
Dependencies: one LLM endpoint & one Messaging platform The LLM can be switched by configuration (the default is the aforementioned Bippin).

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25.10.2026